Orange Condition Classification

Classical Machine Learning vs Transfer Learning

Project Report

Repository: github.com/Hemanth-1510/orange_classification_ml

Scope clarification: The supplied dataset was orange-based rather than avocado-based. This report therefore describes classification of the provided orange labels and does not claim to evaluate avocado ripening stages.

1. Abstract

This project compares an SVM using handcrafted colour and texture features with a pretrained MobileNetV2 model for four orange image labels: *blackspot*, *canker*, *fresh*, and *grenning*. On the recorded held-out test set of 99 images, SVM achieved 97.98% accuracy and 97.73% macro F1. MobileNetV2 achieved 92.93% accuracy and 92.37% macro F1. SVM performed better on this split, but the small test set means the results are preliminary.

2. Objective and dataset

The objective was to build two different modelling approaches, compare them using a separate test set, examine dataset limitations, and provide a single-image inference demo. The original assignment title referred to avocado ripening-stage detection, but the supplied Drive dataset contained orange images; the scope was transparently adapted.

The observed classes were blackspot, canker, fresh, and grenning. The spelling *grenning* is retained from the source folder. These labels combine condition/defect categories and freshness; they do not form a complete ordered ripening-stage scale.

The recorded training folder contained 991 images. A stratified 85/15 training-validation split yielded 842 training images and 149 validation images. The separate held-out test folder contained 99 images.

3. Preprocessing and models

For SVM, images were resized to 128×128. Features consisted of normalized 32-bin histograms for each channel in HSV and LAB colour spaces, plus a uniform Local Binary Pattern (LBP) texture histogram using 24 sampling points and radius 3. The concatenated feature vector had 203 values. A StandardScaler + RBF-kernel SVC (C=10, class_weight='balanced') was used.

For MobileNetV2, images were resized to 224×224 and normalized using ImageNet statistics. A pretrained MobileNetV2 was loaded, its convolutional feature layers were frozen, and its final classifier was replaced with a four-class linear layer. Training used Adam at learning rate 0.001 for five epochs, with horizontal flip, small rotations, and colour jitter augmentation.

4. Held-out test results

Model	Accuracy	Macro precision	Macro recall	Macro F1
SVM	97.98%	97.73%	97.73%	97.73%
MobileNetV2	92.93%	93.01%	92.05%	92.37%

SVM correctly classified 97/99 test images; MobileNetV2 correctly classified 92/99. SVM accuracy was 5.05 percentage points higher and macro F1 was 5.36 points higher.

Class-wise precision / recall / F1 (support)

Model / class	Precision	Recall	F1	Support
SVM — blackspot	0.95	0.95	0.95	22
SVM — canker	0.95	0.95	0.95	22
SVM — fresh	1.00	1.00	1.00	33
SVM — grenning	1.00	1.00	1.00	22
MobileNetV2 — blackspot	0.91	0.91	0.91	22
MobileNetV2 — canker	0.89	0.77	0.83	22
MobileNetV2 — fresh	0.92	1.00	0.96	33
MobileNetV2 — grenning	1.00	1.00	1.00	22

5. Interpretation

SVM outperformed MobileNetV2 on accuracy and macro F1 for this split. MobileNetV2's canker recall was 0.77, indicating that some canker images were missed. The deep model reached 100% validation accuracy in one recorded epoch but

scored 92.93% on the final test set, showing why a separate test set matters.

These results do not show that SVM is universally better than deep learning. A different split, larger dataset, or independent source may change the comparison.

6. Dataset bias and limitations

The 99-image test set is small, and class support is uneven (33 fresh images and 22 in each other class). Lighting, camera, fruit variety, background, distance, disease severity, and image-source differences may not be adequately represented. Models may exploit background/source patterns rather than robust fruit features.

The class labels do not form a complete ripening-stage scale, and the dataset does not match the avocado topic in the original assignment. No production latency, calibrated confidence, or field robustness measurements were recorded. The models should not be described as production-ready.

7. Improvements

Collect more images from independent sources and capture conditions, verify labels with clear class definitions, and use source-aware splits to avoid near-duplicate leakage. Run repeated stratified cross-validation on training data while preserving an untouched test set. Inspect confusion matrices and misclassified canker examples.

Compare SVM feature configurations and tune hyperparameters. For MobileNetV2, try unfreezing selected layers, learning-rate scheduling, and early stopping. Measure inference latency on a documented device and test robustness on a larger external dataset.

8. Demo and reproducibility

The final Colab notebook should include dataset inspection, sample images and class counts, both model implementations, validation and test results, confusion matrices, and a single-image prediction demo. The repository provides scripts for SVM feature extraction/training and a MobileNetV2 training template. Copy the actual trained model files from Colab into *models/*; they are not included in this generated package because they were not available in this chat.

9. Mandatory assignment questions

Q1. How well does the model perform? On the 99-image test set, SVM achieved 97.98% accuracy and 97.73% macro F1, outperforming MobileNetV2 (92.93% accuracy, 92.37% macro F1). This is promising for this split but does not establish real-world performance.

Q2. What are the limitations? The dataset and test set are small and may not cover varied lighting, cameras, backgrounds, fruit varieties, or defect severity. Labels are not a complete ripening scale, and the provided data is orange-based rather than avocado-based.

Q3. What would you improve? Collect and verify more diverse images, improve class definitions, run repeated cross-validation, tune the SVM, partially fine-tune MobileNetV2, and evaluate on a larger independent test set. Also measure inference latency and robustness.

Q4. What took the most time and what did you learn? Dataset inspection, handcrafted feature extraction, training both model types, and comparing validation with held-out test results took the most effort. The main learning was that a classical model can outperform transfer learning on a small image dataset, and validation performance alone can be misleading.

10. Conclusion

The SVM was the strongest recorded model for this orange dataset, reaching 97.98% test accuracy and 97.73% macro F1. Conclusions are limited to this dataset and split. More diverse data and independent evaluation are required before practical use.